

REVIEW ARTICLE

Soil functional responses to ecological restoration treatments in frequent-fire forests of the western United States: a systematic review

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We investigated general effects of ecological restoration treatments on soil function in frequent-fire forests of the western United States using a systematic review methodology. We searched numerous publication databases for original research papers and used well-defined criteria developed a priori to select papers for review. We used meta-analysis and qualitative summaries to compare reported responses of macronutrients, nitrogen cycling, and soil respiration among tree thinning (thin), prescribed fire (burn), and thinning plus prescribed fire treatments (composite). Results of meta-analysis showed that mean differences in macronutrients were consistently higher in composite treatments (standardized using controls) when compared to thin-only and burn-only treatments. Mean responses related to nitrogen cycling showed similar patterns, with significant increases detected in composite treatments for all nitrogen cycling variables (mineralization, ammonification, and nitrification) and insignificant responses for the majority of the burn-only and thin-only treatments. Mean difference in response for soil respiration following composite treatments showed increases as compared to the controls, and no significant differences were detected in response to burn- and thin-only treatments. While soil function, nutrient cycling, and soil respiration differed among treatments, the most significant effects were observed for nitrogen and carbon responses, net mineralization and nitrification, ammonium availability, and soil respiration rate, which experienced the greatest increase following treatments that were both thinned and burned.

Key words: fire, macronutrients, nutrient cycling, *Pinus jeffreyi*, *Pinus ponderosa*, thinning

Implications for Practice

- This study substantiates that systematic reviews are powerful and efficient tools for providing reproducible summaries of evidence relevant to restoration treatments and exhibit great potential to guide policy decisions and consideration of management alternatives.
- The most significant responses to restoration treatments were positive and observed for total nitrogen- and carbon-related variables, ammonium availability, net mineralization and nitrification, and soil respiration rate.
- The most significant responses to restoration treatments resulted from composite treatments that included both tree thinning and prescribed fire.
- Manipulation of forest structure and composition, particularly through thinning and/or prescribed fire, may affect soil processes and function and consideration should be given to the complex nature of interactions among vegetation, soil microbiota, and physicochemical processes when planning restoration treatments.

variability for structure, pattern, and ecological processes (e.g. Burrows 2008; Moya et al. 2011; Franklin & Johnson 2012). However, the reintroduction of fire and the use of prescribed burning, alone or following thinning, remains somewhat contentious among scientists and within the broader community (Clarke 2008; Morrison et al. 1996; Reinhardt et al. 2008; North et al. 2015). Globally, tree mortality as a result of sustained drought, uncharacteristic wildfires, and biotic agents appears to be increasing, driven in part by climate change (van Mantgem et al. 2009; Allen et al. 2010; Weed et al. 2013; Dennison et al. 2014). In some ecosystems, such as the western frequent-fire forests of the United States (including forests dominated by ponderosa pine, *Pinus ponderosa*, and

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Introduction

Many frequent-fire ecosystems around the globe are actively managed for fuel reduction to minimize fire severity, resilience to disturbances, and to reestablishing natural ranges of

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Jeffrey pine, *Pinus jeffreyi*), anthropogenic disruption of natural disturbance regimes has led to large increases in tree and woody plant density, reductions in understory plant community abundance and species richness, and increases in forest floor O horizon (Covington & Moore 1994; Kaye et al. 1999; Laughlin et al. 2006). Currently, overly dense forests show symptoms of ecological degradation in terms of lack of resilience to natural disturbance and loss of habitat quality and natural levels of heterogeneity (Waltz et al. 2014).

Although reference information on soil processes is lacking, substantial alteration of ecosystem function has likely accompanied structural and compositional changes stemming from fire exclusion (Hart et al. 2005; Grady & Hart 2006). For example, a strong body of research indicates that increases in sequestration of nutrients in needle accumulations on the forest floor have resulted in reduced nitrogen transformation and carbon cycling rates (Kaye & Hart 1998; Selmants et al. 2008). Lack of frequent surface fire and associated reduction in abundance of N-fixing plant species may have negatively affected soil C and N fluxes and concentrations in upper horizons (Johnson & Curtis 2001). Increases in forest cover and shading, increases in litter depths, decreases in soil temperature due to shading, and decreased moisture availability due to increased tree densities have also likely negatively impacted microbial community activity and soil respiration (Ma et al. 2004; Concilio et al. 2005; Ryu et al. 2009).

Ecological restoration treatments in these forests assist the recovery of ecosystem function through removal of smaller, younger trees, management of fuel loads, and reintroduction of low-intensity surface fire (Covington et al. 1997; Allen et al. 2002; Fulé et al. 2012). In practice, the success of a restoration project is often based on the establishment of a target plant community rather than on the establishment of a vital, dynamic, and living community of organisms belowground, or the functions of these soil communities, primarily because ecologists tend to focus on soils as a medium for plant growth (Callahan et al. 2008). The impact to the physical, chemical, and biological properties of the soil following ecological restoration treatments is of fundamental importance and interest.

Soil processes in natural systems are complex, and our understanding of these processes in relation to ecological restoration practices is still in relatively early stages. Manipulation of forest density and application of prescribed fire may affect soil function in a number of ways, and responses reflect complex interactions among vegetation, soil microbiota, and physicochemical processes. For example, tree thinning activities that decrease light interception by forest canopies, and increase soil temperature and moisture, may lead to increased microbial activity, respiration, and nutrient cycling (Hart et al. 2006; Selmants et al. 2008). However, whole-tree removal may result in net losses of nutrients and carbon from sites (Johnson & Curtis 2001) while nutrient cycling may remain unaffected where thinning and logging residues are not subsequently burned or removed (DeLuca & Zouhar 2000). While treatments that include prescribed fire have common outcomes such as a net nutrient influx, enrichment of P, Ca²⁺, Mg²⁺, K⁺, and an often temporary increase in pH, treatment effects on soil nutrients and biotic responses are often

variable (sometimes including no net change) and depend upon attributes examined, time frame, intensity of fire, and frequency of burning (DeBano et al. 1998; Hart et al. 2005; DeLuca & Sala 2006). Thus, prediction of soil functional responses to ecological restoration treatments is problematic and to date no systematic analysis of general patterns or drivers of variation in patterns in this area has been undertaken.

Systematic reviews use an evidence-based framework to review the effectiveness of various interventions, including manipulations at an ecosystem level (Pullin & Stewart 2006). Systematic reviews are powerful and efficient for assisting with knowledge transfer and have an aim of providing reproducible summaries of evidence to guide policy decisions and consideration of management alternatives (Roberts et al. 2006; Bilotta et al. 2014). They differ from traditional narrative reviews in a number of ways, from question formulation all the way through to dissemination of the results. For some reviews, prepublished protocols are developed. In addition, members of the team attempt to locate all relevant published and unpublished studies in order to lessen publication bias, and the type of studies to be included are defined prior to the systematic literature search. Reviewers attempt to assess study quality and base conclusions from those papers on sound methodology (Petticrew 2001; Bilotta et al. 2014). Systematic reviews are not a substitute for quality primary research, but they serve a valuable purpose of identifying where primary research may be lacking (Petticrew 2001). Although systematic reviews have been produced in the medical field for decades, many environmental researchers, practitioners, and policymakers are just now becoming familiar with the steps involved in developing such reviews (Petticrew 2001; Bilotta et al. 2014).

Furthermore, it is important to realize that systematic reviews differ from traditional narrative reviews in numerous ways. Narrative reviews tend to be largely qualitative, do not adhere to a strict systematic search of the literature, and often concentrate on a subsample of studies chosen based on convenience or author selection. Thus narrative reviews, while informative, often include selection bias. Additionally, narrative reviews can be confusing, particularly if comparable studies have differing results. Systematic reviews, as the name implies, should be conducted according to a detailed and comprehensive plan and search strategy, with the goal of reducing bias by identifying, appraising, and synthesizing all relevant studies on a particular topic. Often, systematic reviews include a meta-analysis component which uses statistical techniques to synthesize the data from several studies into a single, directly comparable, standardized estimate, or summary of effect sizes (Petticrew & Roberts 2006).

In this study, we reviewed published literature to determine whether ecological restoration treatments in frequent-fire forests of the western United States generally resulted in positive, negative, or neutral effects on soil function. We utilized a systematic review methodology, and due to the availability of data, we chose to focus on response data for three basic indicators: macronutrient stocks, macronutrient cycling, and soil respiration.

Table 1. Inclusion criteria of studies and search terms utilized in the review.

Inclusion Category	Specific Criteria	
Subjects	Western United States (Pacific Northwest, Sierra Nevada, Rocky Mountains, and Southwest) coniferous forests dominated by (1) ponderosa pine (<i>Pinus ponderosa</i>), Jeffrey pine (<i>Pinus jeffreyi</i>), (2) pines mixed with oak (<i>Quercus</i> spp.), or (3) dry mixed conifer forests dominated by one of these pine species but also containing firs (<i>Abies</i> spp.), Douglas-fir (<i>Pseudotsuga menziesii</i>), other pine species (e.g. <i>Pinus lambertiana</i> , <i>Pinus coulteri</i>), and quaking aspen (<i>Populus tremuloides</i>)	
Time frame	1940–present	
Treatments	Treatments conceived as ecological or ecosystem “restoration,” hazardous fuel reduction or ecosystem/forest health management including: thin only; burn only (prescribed fire, controlled fire, and/or wildland fire use); composite (thin and burn); control (untreated); pre- and post-treatment observations	
Outcomes	Soil function and ecological restoration variables: soil respiration; nutrient cycling, macronutrients (N, K, Ca, Mg, P, and S); soil carbon; nitrogen cycling (mineralization, ammonification, and nitrification); microbiota (bacteria, fungi, actinomycete, microbes, and other microfauna); plant root-fungal symbionts (mycorrhizae, ectomycorrhizae)	
Search terms	Soil infiltration Soil moisture Soil temperature Respiration Nutrient availability Macronutrient Nitrogen or Nitrate or Nitrification Potassium Nutrient uptake Plant uptake Phosphorus Nutrient cycling Mineralization or Mineralizable Ammonia or Ammonium or Ammonification Amino Bacteria Fungi or Fungal or Fungus Protozoa Actinomycete	Heterotroph* Microbiota Microflora Microorganism Microbe or Microbial Fermentation Rhizosphere Mycorrhiza* or Ectomycorrhiz* Fine root* Denitrification Microfauna

Methods

We conducted our review following the steps suggested by Pullin and Stewart (2006): (1) formulate questions, (2) develop a review protocol, (3) search data sources, (4) select relevant data, (5) assess the quality of methodology (evidence), (6) extract data, and (7) synthesize (analyze) data. We drafted questions, which were then refined using informal discussions with outside experts in natural resource management. We developed a review protocol that described criteria for data searching, inclusion, and quality of evidence (Table 1).

We systematically searched CAB ABSTRACTS (which includes Forest Science Database), Web of Science, Biosis, and GREENfile for original research papers using the review protocol, search terms, and inclusion criteria provided in Table 1. Searches were conducted on both the common and scientific names of the primary tree species, ponderosa and/or Jeffrey pine, and had to include the terms “soil” and “forest.” Types of literature were limited to primary research and included only refereed journal articles and peer-reviewed scientific reports. We evaluated each study to insure that the results presented were based on unique data and assessed them against our inclusion criteria (Table 1), with only studies considered relevant being included in a database for review and analyses. All studies were

individually reviewed and data were extracted using Engauge Digitizer 5.1 (<http://digitizer.sourceforge.net>) and summarized in a spreadsheet using categories describing the subjects (forest type, geographic region), treatments (burn only, thin only, thin and burn), and outcomes (nitrogen, phosphorus, nitrate, soil respiration, microbiota, etc.) (Table 1).

Quantitative data on soil function and ecological restoration responses were compared across forest types and treatments (e.g. mechanical thinning, prescribed fire; see Table 1 for a full list of treatments considered) using meta-analysis techniques and qualitative summaries. Meta-analysis techniques (Gurevitch et al. 1992) and the metaphor package (Viechtbauer 2010) were implemented in R 3.2.1 (R Core Team 2015). Meta-analyses commonly use standardized mean differences or “effect sizes” (e.g. Cohen’s *d*; Cohen 1969), which are calculated based on sample size and variance, assuming that studies with large sample sizes and smaller variances are more reliable (Hedges & Olkin 1985). For each response with two or more observations, we calculated a standardized mean effect size with 95% confidence intervals and corrected for negative bias, yielding interval estimates of Hedges’ *g* (Hedges 1981). Mean effect sizes were considered to be significantly different from 0 when the corresponding confidence interval did not

include 0. A standardized mean difference of 0, positive, or negative indicated no change between treatment and control, an increase in the response variable compared to the control, or a decrease in the response variable, respectively. For each outcome, meta-analytic fixed-effects models were fit without covariates and using inverse-variance weighting to obtain summary estimates or an overall estimated mean difference (Thompson & Sharp 1999) in effect size.

To examine the robustness of our study against publication bias (i.e. the “file drawer problem” or the failure to publish null results), we computed “fail-safe” numbers (Rosenthal 1979; Gates 2002) for effect size (g) estimates. A fail-safe number estimates the number of additional null-result studies needed to produce statistically insignificant results, or the higher the number the more confident we are in detected effects. The reliance of this technique on variance estimates limited fail-safe number estimation to 35 of the 43 studies only.

In cases where a measure of variability (e.g. standard deviation) between treatment means was not reported and each response contained two or more observations, we calculated a variation of a response ratio (Hedges et al. 1999), relativized to the control mean, as our measure of effect size. We defined this standardized response as:

$$\overline{SR} = \left(\frac{\text{Control mean} - \text{treatment mean}}{\text{Control mean}} \right)$$

Response ratios have become commonly used in meta-analysis (Schmitz et al. 2000; Mosquera et al. 2000; Côté et al. 2001; Ainsworth & Long 2005; Parker et al. 2006; Miller et al. 2010; Fulé et al. 2012), as they are specifically designed to measure relative differences. For each outcome or response, the mean standardized response and standard deviation were calculated. In cases where standardized mean differences or standardized response could not be calculated given the information reported, qualitative descriptions of responses were summarized.

Results

Our initial database search returned 942 candidate papers. After abstract and title review, we retained 110 papers to evaluate for inclusion based on the criteria presented in Table 1. Forty-nine met inclusion criteria but three of these were excluded from review ($n = 46$) because reported outcomes were strictly expressed in terms of surface fuels (e.g. litter weight) or soil macrobiota (e.g. density of mites). Studies were located throughout the western United States and spanned a range of frequent-fire forest types, with northern Arizona (42%), California's Sierra Nevada Mountains (21%), and Oregon (14%) being the most strongly represented (Fig. 1). Least represented (making up approximately 19% of the data) and in decreasing order, were studies from Montana, Nevada, New Mexico, Idaho, and Colorado (with the latter two states only including one study from each). Thirty-six studies (75%) were published in the year 2000 or later. Just over half (55%) of the studies were from ponderosa pine forest types, with the remaining

studies being conducted in mixed conifer forests dominated by either ponderosa (30%) or Jeffrey (15%) pine. Study sites ranged from 1,150 (Gundale et al. 2005) to 2,590 m (Concilio et al. 2005) in elevation.

Six variables related to macronutrients and three related to nitrogen cycling and soil respiration had sufficient data (two or more observations) for meta-analysis. While five studies reported responses related to microbiota (namely mycorrhizal symbionts), no two studies reported common outcomes and therefore none were included in meta-analysis. Macronutrient variables included those related to carbon (C), total nitrogen (N_{total}), available nitrogen (N_{avail}), nitrate (NO_3^-), ammonium (NH_4^+), phosphorus (P), and potassium (K^+); while nitrogen cycling variables included measures of net mineralization (conversion of organic N to mineral forms; hereafter “mineralization”), ammonification (conversion of organic N to ammonium, hereafter “ammonification”), and nitrification (conversion of ammonium to nitrate). Figures 2A and 2B show standardized mean differences in total and available nitrogen responses for thin, burn, and composite restoration treatments. For the studies reporting burn-only treatments for N_{total} ($n = 6$) and N_{avail} ($n = 3$), the mean difference was largely evenly distributed among possible outcomes (e.g. N_{total} exhibited two with increases, two with decreases, and two with no significant effect) resulting in an overall estimated mean difference not significantly different from 0 for both N_{total} and N_{avail} (Fig. 2; -0.52 to 0.36 and -0.46 to 2.18 , respectively). Similar to the burn-only results, overall estimated mean difference for N_{avail} following thin-only treatments was not significantly different from 0 (-0.18 to 1.57) and was slightly negative (-1.19 to -0.19) for N_{total} . All studies reporting effects of composite treatments on N_{total} and N_{avail} showed positive mean increases with eight of ten being significantly different from 0. Overall estimated mean differences for N_{total} and N_{avail} were 3.73 and 2.91 , with 95% confidence interval estimates of 3.0 – 4.47 and 1.97 – 3.85 , respectively.

Figure 3 demonstrates 95% confidence intervals of overall estimated mean difference summarized for all macronutrients (including N_{total} and N_{avail}). Mean difference in macronutrients showed a consistent pattern of significant increases compared to controls in composite treatments, while burn and thin treatments alone were lower in difference as compared to the untreated controls. One exception in this pattern was mean difference in NH_4^+ (Fig. 3D), where thin-only treatments resulted in a 4.52 – 9.66 increase. Burn treatments significantly decreased C and NH_4^+ , and had no detectable effect on the remaining macronutrients. Thin treatments significantly decreased P and N_{total} , increased NH_4^+ and K^+ , and had no significant effect on remaining macronutrients.

Estimated mean difference for variables related to nitrogen cycling showed a similar pattern to those observed for macronutrients. Specifically, results (Fig. 4A–C) suggest significant increases compared to controls in composite treatments for all nitrogen cycling variables (mineralization, ammonification, and nitrification), and insignificant responses for all burn- and thin-only treatments, except for increased mineralization observed for thin treatments (Fig. 4A). While mean difference

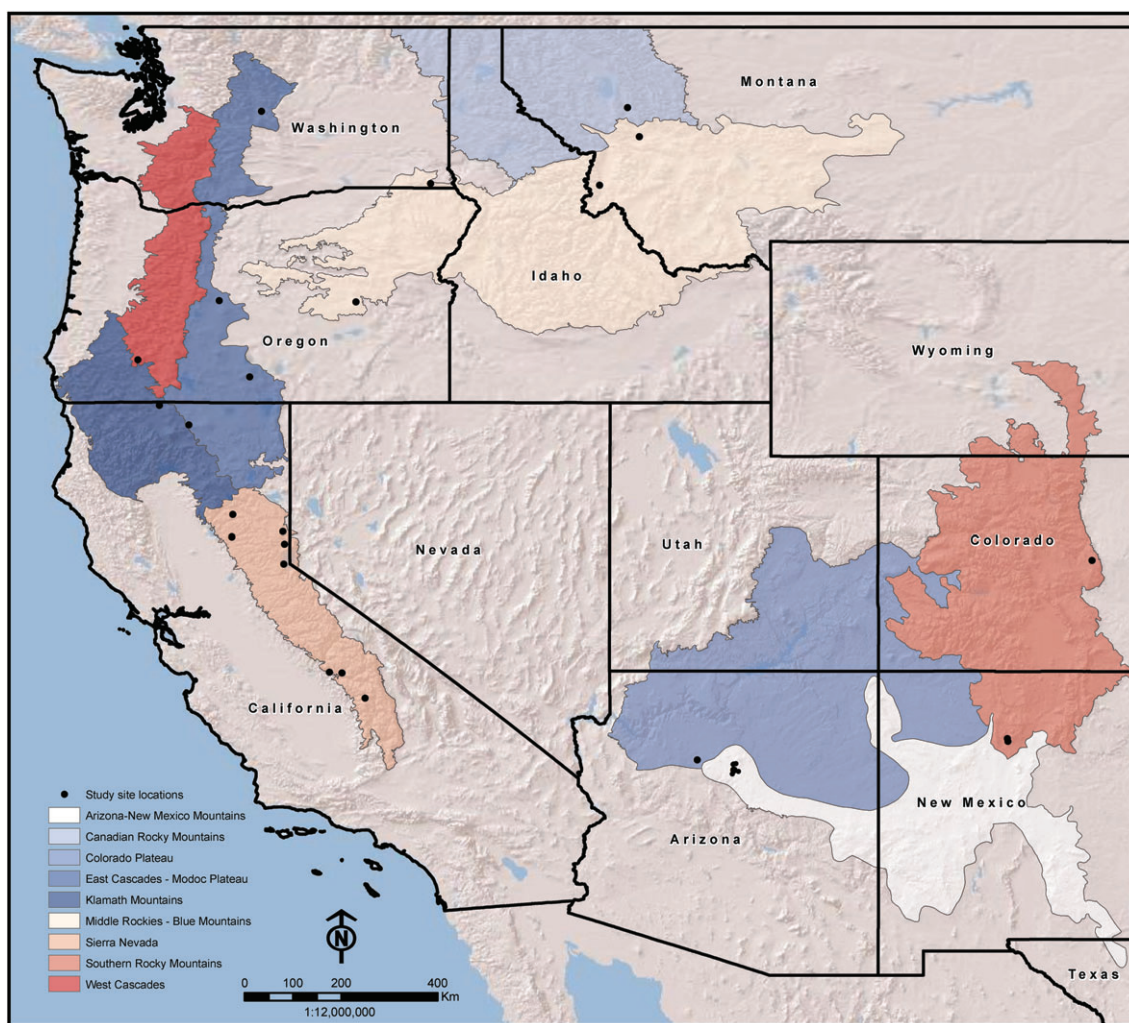


Figure 1. Geographical regions of the western United States containing coniferous forest dominated by ponderosa pine (*Pinus ponderosa*), Jeffrey pine (*Pinus jeffreyi*), or dry mixed conifer forests dominated by one of these pine species and approximate locations of studies included in the review (Appendix S1, Supporting Information).

in response for soil respiration ranged from 0.72 to 1.45, and averaged 1.09 as compared to untreated controls, no significant difference from controls was detected for burn- and thin-only treatments (Fig. 4D).

Given the number of studies available for meta-analysis, fail-safe numbers were relatively low for all response variables when examining burn- and thin-only treatments (Figs. 3 & 4). In contrast, fail-safe numbers for composite treatments were relatively high for all response variables, and ranged from 21 for potassium (Fig. 3G) to 221 for N_{total} (Fig. 3b). However, ammonification had a composite treatment fail-safe number of 3 (Fig. 4B).

Standardized macronutrient responses, carbon and nitrogen cycling-related responses, and soil respiration for three restoration treatments had inconclusive results (Table 2). With the exception of a significant increase in carbon-related responses ($\overline{SR} = 0.93$, $SD = 0.07$) and an insignificant decrease in nitrification-related responses ($\overline{SR} = -0.87$, $SD = 1.17$), all

other responses to burn-only treatments exhibited significant decreases. Conversely, all responses to thin-only and composite treatments were not significantly different from 0, with the exception of a decrease in carbon-related responses following composite treatments ($\overline{SR} = -0.14$, $SD = 0.15$) and an increase in ammonium-related responses following thin-only treatments ($\overline{SR} = 0.23$, $SD = 1.09$).

Discussion

Fisher and Binkley (2000) refer to soils as the “hidden half” of forests; a half which may be of equal importance to that aboveground, but which is more difficult to observe and therefore less frequently studied. Abundant research suggests that the removal of fire from frequent-fire forest types in the western United States results in various types of deviation from reference conditions observable aboveground, including: (1) increased fuel loads that lead to an infrequent stand-replacing fire regime

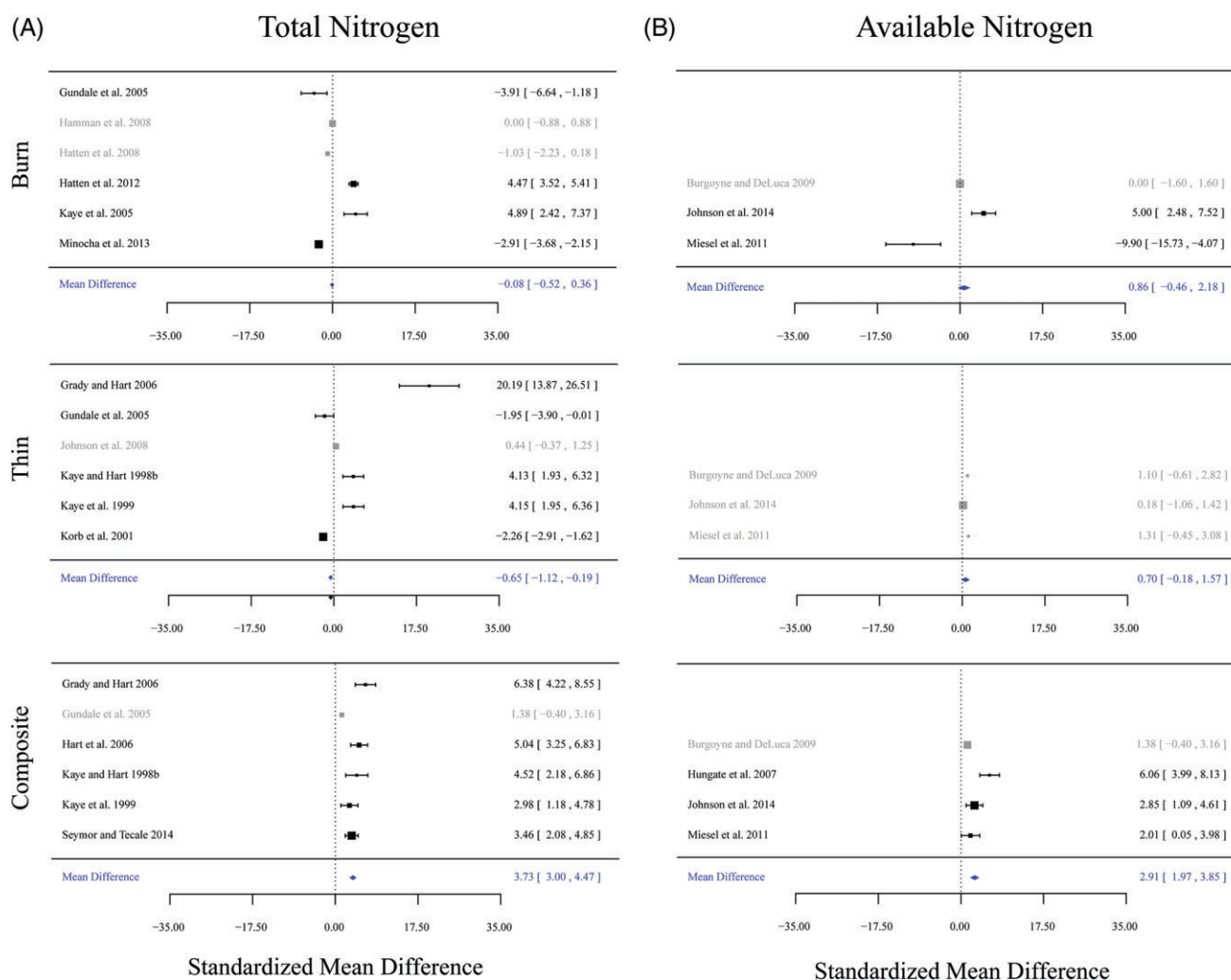


Figure 2. Forest plot showing standardized mean differences in total (A) and available (B) nitrogen responses for thin, burn, and composite restoration treatments. Positive mean difference values indicate nitrogen increases, whereas negative difference indicates decreases. Responses were considered not statistically significant (grayed) when confidence intervals intersected 0.00 difference (dotted line).

(Kalies & Yocom Kent 2016); (2) reduced biodiversity and loss of herbaceous plants (Moore et al. 2006); and (3) decreased productivity in the form of tree and stand growth due to overcrowding (Kolb et al. 2007). Effects resulting from long-term absence of fire may also be evident belowground due to indirect effects of high tree biomass and concomitant litter deposition and accumulation. Higher tree densities may induce lower water availability (Kaye et al. 1999) due to consumption and canopy interception, possibly slowing microbial activity and decomposition (Aerts 1997). Litter accumulation may further alter nutrient cycles because tree litter, particularly that of conifers, tends to be of low quality (i.e. high lignin:N and C:N), suppressing decomposition rates (Aerts 1997) and favoring nitrogen immobilization over mineralization (Lewis & Kaye 2012). Both enhanced canopy and litter cover can be expected to cool the soil during the growing season, which may also suppress microbial activity, and therefore decomposition, due to a strong positive dependence on temperature (Davidson & Janssens 2006). Our

systematic review sought to determine if the results from ecological restoration studies designed to restore aboveground structure, composition, and function could provide evidence of soil functional outcomes.

Restoration Treatments Effect on Soil Function

The response of physical, chemical, and biological soil properties to forest restoration treatments is fundamentally important because soils are the sites where most decomposition occurs and are thus directly linked to the nutrient cycling process. Decomposition, along with primary production, is one of the two major life-generating processes on Earth (Wall et al. 2005), and mineral nutrient pools may constrain production (Wieder et al. 2015). As ecological restoration seeks to assist the recovery of normal ecosystem function (SER 2004), achieving normal decomposition and nutrient cycling rates are integral to the practice, though often not explicitly addressed. Our meta-analysis of macronutrients, nutrient cycling, and soil

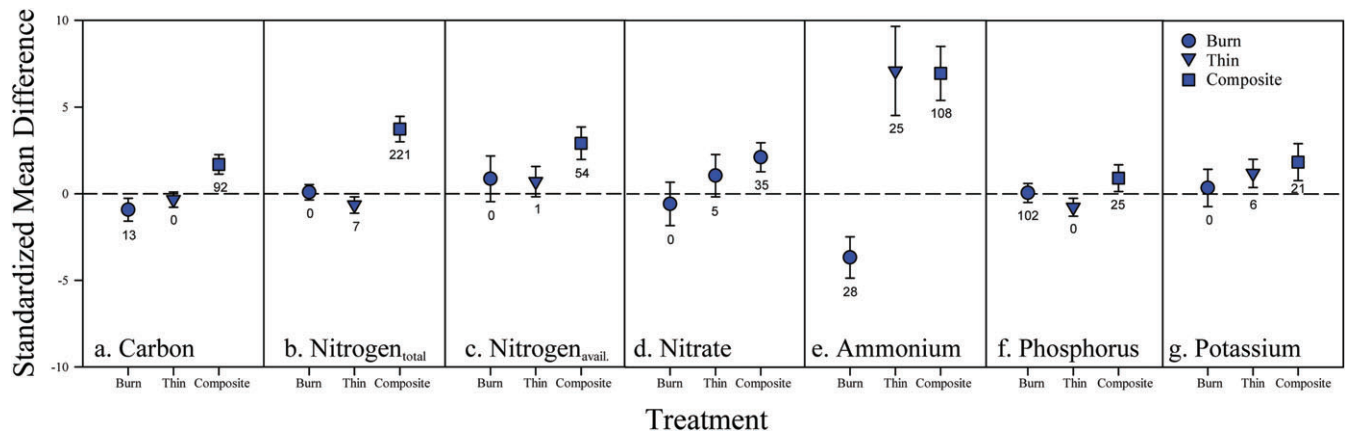


Figure 3. Standardized mean difference in effect size and $\pm 95\%$ CI for three restoration treatments compared for each of six macronutrient-related responses. Standardized mean difference of 0, positive, or negative indicated no change between treatment and control, increases in the response variable compared to the control, or decreases in the response variable, respectively. The number below each confidence interval is the corresponding “fail-safe” number (Rosenthal 1979) for effect size.

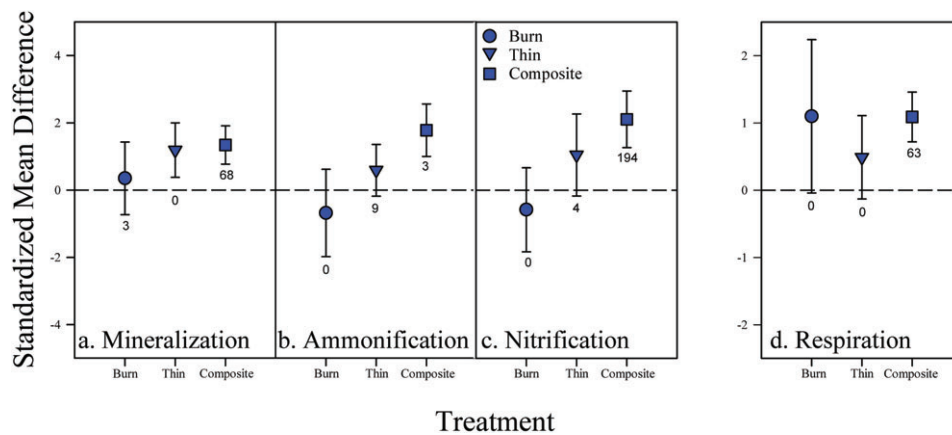


Figure 4. Standardized mean difference in effect size and $\pm 95\%$ CI for three restoration treatments compared for each of three nitrogen cycling-related responses and for soil respiration-related responses. Standardized mean difference of 0, positive, or negative indicated no change between treatment and control, increases in the response variable compared to the control, or decreases in the response variable, respectively. The number below each confidence interval is the corresponding “fail-safe” number (Rosenthal 1979) for effect size.

respiration responses to forest restoration treatments across the western United States indicated that ecological restoration treatments implemented aboveground can effectively alter soil function belowground, but that various techniques are differentially effective.

The impacts of fire on soils depend strongly upon the heat generated, which is in turn well-predicted by the amount of fuel consumed (DeBano et al. 1998). In our analysis, burning alone either significantly reduced or had little short-term effect (≤ 5 years) on most macronutrients, nutrient cycling, and soil respiration. It is very likely that nutrients were lost in the O horizon (Covington & Sackett 1986), but such an effect did not translate to mineral horizons or undifferentiated surface soils. These trends are likely the result of multiple processes. Carbon on or near the soil surface is lost due to combustion, but ash and other residues are also a C input. This may explain some apparently conflicting results suggesting burning alone may either increase or decrease soil C. Nitrogen is susceptible

to loss through oxidation, volatilization, or convection in ash (Fisher & Binkley 2000). Organic P may also volatilize, but at higher temperatures. In contrast, metal cations such as K^+ , Ca^{2+} , and Mg^{2+} neither volatilize nor oxidize; rather they accumulate in ash fall (Fisher & Binkley 2000). Although a burned stand experiences a net loss of nutrients, ash fall represents a nutrient input into soil pools from aboveground pools or from O horizon to mineral soil horizons, at least initially. The concentration of K^+ , Ca^{2+} , and Mg^{2+} ions in the soil can increase or be unaffected by fires, whereas nitrogen and sulfur often decrease (Hough 1981). Additionally, NH_4^+ is a direct product of the combustion of organic matter, while NO_3^- is created from NH_4^+ sometime after fire as a result of nitrification (Covington & Sackett 1992). If not promptly taken up by plants or microbes (both of which are likely reduced by the fire), available nutrients including PO_4^{3-} , K^+ , Mg^{2+} , Ca^{2+} , NH_4^+ , and NO_3^- can quickly decrease, either by leaching (e.g. during snowmelt events), erosion (e.g. during summer monsoonal storms), or sorption reactions that

Table 2. Summary attributes (mean, standard deviation, and sample size) for standardized responses ($\overline{SR}$) examined for three restoration treatments compared for each of nine macronutrient-related responses (top) expressed per unit area or mass and for soil respiration, carbon, and nitrogen cycling-related responses (bottom) expressed as a rate. Standardized responses reflect variability in the short term (≤ 5 years since treatment), site variability, measurement unit, treatment differences, and individual study sample size effects. Values considered significantly different from 0 (bold) when their corresponding $\pm 95\%$ CI did not include 0.

	Burn			Thin			Composite		
	Mean	SD	n	Mean	SD	n	Mean	SD	n
SR per unit area or mass									
C	0.93	0.07	12	−0.03	0.30	15	−0.14	0.15	14
Ca	−0.91	0.08	5	−0.15	0.29	5	0.15	0.29	8
K	−0.72	0.47	5	0.10	1.52	5	−0.11	0.21	7
Mg	−0.94	0.07	5	−0.21	0.32	5	−0.14	0.13	4
N _{avail}	−0.75	0.40	7	0.26	0.74	7	0.52	1.55	8
N _{total}	−0.76	0.56	12	0.18	0.78	16	−0.02	0.16	15
NH ₄	−0.77	0.64	12	0.23	1.09	16	0.29	2.94	23
NO ₃	−0.69	0.40	10	0.00	0.91	13	2.22	277.36	21
P	−0.85	0.25	13	0.02	2.33	14	0.09	0.45	10
SO	−0.85	0.17	3	−0.11	0.40	3	Insufficient data		
SR expressed as a rate (per unit area or mass per time period)									
C	−0.95	0.00	2	−0.06	0.05	2	−0.05	0.04	2
N _{avail}	−0.73	0.32	6	0.3	0.94	6	0.19	1.08	7
NH ₄	−0.64	0.27	4	0.2	0.74	4	0.02	0.39	4
NO ₃	−0.87	1.17	6	0.99	2.36	6	1.50	2.46	5
Respiration	−0.83	0.11	10	−0.04	0.30	15	0.14	0.43	14

render nutrients less available (Mroz et al. 1980; Kaye et al. 1999; Johnson & Curtis 2001).

Thinning is likely to affect soils in fundamentally different ways. Rather than directly impacting nutrient pools, removal of trees allows more solar energy to reach the soil surface, and may favor greater soil moisture due to decreased evapotranspiration and canopy interception of precipitation (Sun et al. 2015). Because this alteration of the soil environment affects microbial activity, thinning activities may result in increased soil function, respiration, and nutrient cycling (Hart et al. 2006; Selmants et al. 2008). Recent studies indicate that a positive response of nutrient concentrations or soil respiration to forest thinning is not universal. For example, Klemmedson (1992) reported little or no loss in N, C, and S and significant increases in P and K following thinning in ponderosa pine forest of northern Arizona, while Tang et al. (2005) report that thinning decreased soil respiration as a result of decreased root density after thinning in a ponderosa pine plantation in the Sierra Nevada Mountains. However, soil function may remain unaffected in stands where tree removal slash is left in place and not burned (DeLuca & Zouhar 2000). Our results suggest that it is difficult to detect an overall effect of thinning on soil nutrient pools, cycling or respiration, likely due to such variability among study sites. A notable exception was that thinning alone appeared to have a positive effect on nitrogen cycling, especially the NH₄⁺ pool.

Composite treatments that included tree thinning and prescribed fire generally induced showed greater positive soil responses compared to burning or thinning alone, a finding similar to that reported for fuels reduction treatments by Kalies and Yocom Kent (2016). This was true for all of the measured processes, and several nutrient pools. We believe that this is either because less fuel was consumed in the cases where debris

was removed before burning, or because less of the soil surface was exposed to high temperatures in the cases where pile burns were implemented. Lower burn temperatures should lead to fewer nutrient losses by volatilization, oxidation, or convection (Fisher & Binkley 2000). Concentrating the impact of burning, and therefore omitting burn impacts over much of the soil surface, is likely to retain more of the plant cover, and soil biota biomass, thus reducing leaching losses. While burning alone has primarily negative effects on soil nutrient pools and cycling, and thinning has some positive effects, the composite treatment illustrates an interactive effect on frequent-fire forest ecosystems. Like treatments employing burning alone, losses of nutrients from O-horizons almost surely occurred in composite treatments compared to thinning-only treatments. However, the presence of fire is an explicit part of the desired outcome of restoration of these ecosystems, and fire alleviates the need for costly recurrent thinning and removal of biomass. We suggest that, of the three ecological restoration approaches, composite treatments most closely mimic the evolutionary environment of these ecosystems (Moore et al. 1999), and therefore is most likely to produce favorable responses both aboveground and belowground.

While our systematic review suggests that composite treatments may exhibit an additive effect on macronutrient stocks, macronutrient cycling, and soil respiration, limited understanding exists for equivalence among contrasting treatments (e.g. burning alone vs. thinning and burning) in terms of soil and ecosystem function worldwide. Simply speaking, the effects of prescribed burning alone on soil function are better understood compared to those resulting from burning following tree thinning or harvesting. For example, Wang et al. (2012)

reported significant decreases in soil organic C and soil respiration, with increases in soil total N (observed relative to the control) in a meta-analysis of 76 papers related to wildfire, prescribed burning, and soil macronutrients from around the globe. Prober et al. (2005) observed similar declines in soil ammonium and few changes in nitrate following repeated prescribed fires, concluding that soil nitrate fluctuated greatly following burn-only treatments in restored grassy woodlands in temperate agricultural regions of Australia. Banning et al. (2007) found that thinning affected soil N cycling up to at least 2 years post-treatment, by increasing the potential for nitrification, particularly when combined with prescription burning for rehabilitated post-mining forests in Western Australia. Furthermore, our results align well with numerous studies reporting inconclusive or net losses in soil macronutrients following single and multiple prescribed burns in fire-prone grassy woodlands of Australia (Prober et al. 2005; Burrows et al. 2010), Aleppo pine (*Pinus halepensis*) shrublands of Spain (Outeiro et al. 2008), and following tree thinning and prescribed burning in other fire adapted ecosystems of North America (Knoepp et al. 2004; Boerner & Brinkman 2004; Trammell et al. 2004; Grady & Hart 2006; Ganzlin et al. 2016).

Systematic Review Challenges and Data Gaps

The results of our systematic review are representative of a methodical search of published research concerning soils responses to restoration treatments in frequent fire ecosystems of the western United States. Paralleling our findings, similar reviews have noted geographic irregularities in restoration science. For example, just under 80% of the studies used by Martinson and Omi (2013) were from the western United States (initial search was not limited to the United States) of which 50% were studies including data from only California, Arizona, Oregon, New Mexico, and Colorado. Likewise, Kalies and Yocom Kent's (2016) study was dominated by results from California, Arizona, Oregon, and Washington. These studies, with the addition of our work, suggest a possible under-representation of research results in other frequent-fire ecosystems of the western United States (those not dominated by ponderosa pine or Jeffrey pine), such as those in Idaho, Montana, Colorado, and Nevada.

Furthermore, systematic reviews offer many advantages over other research synthesis techniques often used in ecology, forestry, and restoration ecology. The results may be more accurately evaluated, investigated, and updated. These are some of the reasons for its popularity in medicine and other fields. One of the ways in which bias is reduced in other fields is that systematic reviews may be registered before being conducted, with strict protocols detailed in advance and followed during the review process (e.g. Higgins & Green 2009). There have been suggestions to adopt this practice in the field of conservation biology (e.g., Stewart 2010) and while it has slowly become a more prominent method, the fields of ecology, forestry and restoration ecology have yet to adopt them.

While fire and thinning effects on soils have been summarized in review or meta-analysis publications (e.g. see Neary et al.

1999; Johnson & Curtis 2001; Carter & Foster 2004; Certini 2005; Nave et al. 2011), we present the first known application of systematic review methodology to quantitatively assess restoration treatment effectiveness. Nevertheless, our efforts to identify ubiquitous macronutrient, nutrient cycling, and soil respiration response patterns were impeded by numerous obstacles. The primary problems included: (1) inconsistent quality of published data and results; (2) lack of rigorous study designs; and (3) a dearth of replicated studies where "restoration" was an explicit focus. These limitations indicate a clear need for more statistically rigorous studies (i.e. highly replicated, short- and long-term experimental designs with control treatments) and improved data reporting. Failure to sample or report on control treatments as well as collect and/or report quantitative macronutrient, soil respiration, and microbiota responses resulted in rendering numerous studies unusable for meta-analysis.

Although we were able to conduct analyses relevant to macronutrient pools, and N and C cycling, our understanding is still only partial. For example, we lack sufficient information on other N cycling processes such as N-fixation and denitrification. We were also unable to provide any new general insights regarding the soil biota, despite using several search terms intended to capture information about them. The soil biota may change in predictable ways in response to restoration treatments, and these community shifts could have contributed to observed changes in soil function. The major barrier that we encountered was the small number of studies presenting data describing the biota, and wide variation in the data reported among the studies. This is unsurprising, considering that there are many different attributes of the biotic community that could be measured, and many studies simply have different goals. Nonetheless, we can suggest some characteristics of the soil community and their functionality that may respond to ecological restoration and would be of broad interest and utility: (1) the ratio of fungi to bacteria generally declines as organic substrate quality increases (Blagodatskaya & Anderson 1998) and (2) the ratio of arbuscular to ectomycorrhizal fungi would be expected to increase in response to any ecological restoration technique that bolstered herbaceous vegetation. Such a shift in the foundational symbiotic fungi is likely to ripple throughout the soil community, affecting ecosystem function (Antoninka et al. 2008; Cavnagaro et al. 2007). Increasingly economical next generation molecular approaches such as HiSeq2000 and MiSeq Illumina platform sequencing (Caporaso et al. 2012), and bioinformatics pipelines such as QIIME (Quantitative Insights Into Microbial Ecology) (Caporaso et al. 2010), are expanding the possibilities of examining the structure and diversity of soil microbial communities using standardized approaches.

Conclusion

The overall results of our systematic review of functional responses to restoration treatments in frequent-fire forests of the western United States indicate that composite treatments combining small-diameter tree thinning and prescribed fire have an interactive effect and, for most variables measured, produce

greater positive soil responses than burning or thinning treatments alone. When planning ecological restoration projects, in addition to the more well-known impacts to the aboveground vegetation from thinning and burning, the chemical, biological, and physical impacts to soils should be considered of equal importance.

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Supporting Information

The following information may be found in the online version of this article:

Appendix S1. List of systematic review references.

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